

Tristan Johnston

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Full-stack software engineer with production experience building scalable web applications using React, Next.js, TypeScript, Node.js, and PostgreSQL. Shipped features for a platform serving 12,000+ users and built end-to-end systems from database design through frontend delivery.

EXPERIENCE

Software Engineer

Feb. 2026 – Present

Progressive Victory (Civic Tech Nonprofit)

Remote

- Developed production features for a Next.js/TypeScript platform serving a 12,000+ member community, collaborating through GitHub Actions CI/CD, code review, and Scrum.
- Owned features end-to-end, implementing REST API endpoints, MySQL schemas and stored procedures to improve volunteer coordination and administrative capabilities
- Improved backend data handling for edge-case inputs and wrote unit tests, reducing runtime errors and preventing bug introduction in production

Full-Stack Software Engineer

Oct. 2024 – Nov. 2025

A Pet's Day Out

Birmingham, AL

- Architected a PERN (PostgreSQL, Express, React, Node.js) full-stack business dashboard managing 10,000+ pet records with attendance tracking, completion tracking, and record retention, reducing intake errors and improving processing time for 30–40 daily appointments
- Built a 25-endpoint RESTful API implementing CRUD operations and fuzzy search with pagination, eliminating N+1 queries via eager loading
- Designed a relational PostgreSQL schema supporting many-to-many customer–pet relationships and automated household grouping, simplifying management of shared ownership
- Implemented secure authentication with Passport.js, Google OAuth, and httpOnly sessions
- Built a secure photo upload pipeline using AWS S3 pre-signed URLs and temporary object staging using AWS Lambda, preventing orphaned uploads and reducing server load
- Delivered a mobile-first React application, allowing non-technical staff more flexibility on the job

PROJECTS

RAG PDF Search | *Next.js, FastAPI, PostgreSQL, pgvector, OpenAI API*

- Built a Retrieval-Augmented Generation system enabling natural-language search across hundreds of PDF pages
- Implemented a search pipeline using sentence-transformer embeddings and pgvector for fast semantic queries
- Developed microservices with FastAPI and Express to handle automated document parsing, embedding, and storage
- Integrated OpenAI API for context-aware response generation with referenced citations and page numbers

Birmingham Parcel Data Visualization | *Next.js, React, PostGIS, Deck.gl, Docker*

- Built an interactive 3D geospatial visualization of value-per-acre parcel data using Deck.gl and MapLibre to highlight municipal land usage efficiency
- Wrangled, processed, and standardized municipal parcel data using Python scripts and manual validation
- Optimized interactions with datasets of 50k+ parcels using spatial index queries in a Dockerized PostGIS database
- Implemented dynamic color interpolation and view-state clamping for clear navigation across urban parcel data

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, Java, C++

Frontend: React, Next.js, Tailwind CSS, Chakra UI, GSAP, CSS, HTML

Backend: Node.js, Express, FastAPI, Flask, Fastify

Databases: PostgreSQL, MySQL, PostGIS, MongoDB, pgvector

Cloud & Tools: AWS S3, AWS Lambda, Docker, Git, GitHub Actions, Vitest

EDUCATION

University of Alabama at Birmingham

Aug. 2019 – Dec. 2023

Bachelor of Arts in Computer Science, Minor in History